- Low-income high-spenders may value discounts and loyalty programs.

3. **Visual Analysis:**
    - Visualizing clusters helps businesses identify patterns and outliers, making segmentation more intuitive and actionable.

Advanced Considerations

1. **Choosing the Optimal Number of Clusters:** Use the Elbow Method or Silhouette Analysis to determine the best number of clusters.

```
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42) kmeans.fit(scaled_data)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss)
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

2. **Feature Engineering:** Additional features such as purchase frequency or product preferences can enhance segmentation.
3. **Evaluation Metrics:** Use metrics like Silhouette Score to evaluate the quality of clustering:

from sklearn.metrics import silhouette_score score = silhouette_score(scaled_data, kmeans.labels_) print("Silhouette Score:", score) Customer segmentation using K-Means clustering empowers businesses to understand their customers better and make data-driven decisions. By leveraging demographic and behavioral data, companies can craft personalized marketing campaigns, improve customer satisfaction, and drive revenue growth. This example illustrates the power of unsupervised learning in uncovering hidden patterns and creating actionable insights in the business world.